t3-w09 — Need for statistic — Need for statistics: purposes, and the collection and tabulation of data

Run `c0c3-deepseek-native` · conditions C0, C1, C2, C3 · the pack shown is draw 3 of each.

HER LESSON PLAN

Need for statistic — Need for statistics: purposes, and the collection and tabulation of data

Objectives

1. explain the purpose of statistic
2. analyse and interpret statistical data presented in tables

Step 1 — intro: what data is (activation)

Teacher asks questions such as

- **teacher** asks questions such as how many boys and how many girls are in the class
- **student** answer, and the answers obtained are examples of data

Step 2 — the purpose and meaning of statistics

Teacher introduces the concept of statistics as a method of collecting and organising information.

- **teacher** introduces the concept of statistics as a method of collecting and organising information

Step 3 — collecting, tallying and tabulating data

Teacher takes the students through collecting data, recording it with tally marks, and setting it out in a frequency table.

- **teacher** takes the students through collecting data, recording it with tally marks, and setting it out in a frequency table

Evaluation

Assembled from the plan's own **Student Activity**, which between its three items covers both retained objectives. Her Evaluation line sets work from the textbook by reference — "Exercise 17a No 3 and 4 page 158 NGM" — so that question text is nowhere in the plan and cannot be coded. Her lettered markers are kept; nothing is renumbered and no wording is changed. - (a) Define statistic - (b) State three methods of collecting data. - (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

1. (a) Define statistic
2. (b) State three methods of collecting data.
3. (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

HER LESSON NOTE

Verbatim, as transcribed. **C2 and C3 receive this; C0 and C1 never do** — which is the whole of what separates them.

Step 1 — intro: what data is (activation)

```
- (a) How many boys are in this class
- (b) How many girls are in this class
- (c) What is the favourite colour of students in the class.

The answer obtained are examples of data.
```

Step 2 — the purpose and meaning of statistics

```
statistics is the science of collecting, organizing, presenting, analyzing and interpreting
data.

#### Meaning of Data

Data are facts or information collected for a particular purpose.

#### Importance of statistics

It helps us to

- (a) Collect Information
- (b) organize Information
- (c) Compare Information
- (d) Make decisions
- (e) Present Information Clearly.
```

Step 3 — collecting, tallying and tabulating data

```
#### Collection of Data

Data can be collected through

1. observation
2. counting
3. Interview
4. Questionnaire.

#### Tally Marks

Tally Marks are used to record data as they are collected.

**Example:** The fig below show the same data collected in two different ways by two student

(a) bus, car, car, car, lorry, bicycle, bicycle, car, car, lorry, bicycle, car.

The above data can be represented in Tally

| Vehicles | Tally | Total |
|---|---|---|
| car | \|\|\|\|\|/ \| | 6 |
| bus | \| | 1 |
```

```
| lorry | \|\| | 2 |
| bicycle | \|\|\| | 3 |
| Motor Bike | | |

#### Frequency Table

A frequency table is a table showing how many times each item occurs in a set of data.

**Examples:**

Favourite colours of student:

| colour | frequency |
|---|---|
| Red | 18 |
| Blue | 20 |
| Green | 14 |
| Yellow | 16 |
| White | 15 |
| Pink | 30 |
```

C0 `The curriculum objective alone. No lesson plan, no note.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Need for statistic

`subtopic`

Need for statistics: purposes, and the collection and tabulation of data

`term`

Third Term

`week`

9

`lesson_objectives`

• explain the purpose of statistic
• analyse and interpret statistical data presented in tables

Objective 1 — explain the purpose of statistic

`objective_text`

explain the purpose of statistic

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Objective 2 — analyse and interpret statistical data presented in tables

`objective_text`

analyse and interpret statistical data presented in tables

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, current_step, instructional_events, student_tasks, activities, assessment_points, source_evidence, source_figures, objective_learner_state`.

Output — the pack it produced (draw 3)

Why We Count: The Purpose of Statistics

EXPLANATION

Statistics is the branch of mathematics that deals with collecting, organising, presenting, and making sense of information. That information, gathered in the form of facts and numbers, is called data. When you hear "statistics" you might think of long lists of numbers, but the real point of statistics is not just to collect numbers — it is to use them to answer questions and make decisions.

So what is the purpose of statistics? Here are the main jobs it does for us:

To collect information in an organised way. Instead of a jumbled pile of facts, statistics helps us gather data carefully and record it neatly, for example in a table.
To summarise and present data clearly. A long list of numbers is hard to understand. Tables, tallies and charts turn that list into something you can read at a glance.
To make comparisons. With data organised, we can compare one group with another — more boys than girls, more rain in July than in January, and so on.
To spot patterns and trends. Does something increase or decrease? Statistics shows us the pattern behind the numbers.
To help us make decisions and predictions. A shop owner decides how much stock to buy by studying past sales. A farmer decides which crop to plant by looking at rainfall data. A school decides how many buses to hire by counting its students.

When we want to answer a question with data, we go through a simple process. First we collect the data, then we tabulate it (put it into a table so it is neat and organised), and then we read and interpret the table to draw a conclusion. Reading a table means looking at what the numbers actually tell us — not just copying them out.</p>

Let's see this whole process working on a real example.

WORKED_EXAMPLE

Worked example 1
A class of 30 students is asked, "What is your favourite sport?" The teacher writes down every answer: football, football, tennis, football, swimming, tennis, football, swimming, football, tennis, football, football, swimming, tennis, football, football, tennis, swimming, football, tennis, football, swimming, tennis, football, football, tennis, football, swimming, tennis, football.

Purpose: We want to know which sport the class likes most, so we collect the data and tabulate it.

Step 1 — Count each sport (tally) and record it in a table:

Sport	Number of students
Football	14
Tennis	8
Swimming	8
Total	30

Step 2 — Read and interpret the table. The table shows that football was chosen most often (14 students), while tennis and swimming each got 8. So we can conclude that football is the most popular sport in this class. This conclusion came from organising the data — that is the purpose of statistics at work.

WORKED_EXAMPLE

Worked example 2
A shopkeeper records the number of drinks she sells each day of one week, then tabulates the results.

Day	Drinks sold
Monday	40
Tuesday	55

Wednesday	60
Thursday	65
Friday	90

Purpose: The shopkeeper wants to know when most drinks are sold so she can stock up at the right time.

Reading and interpreting the table: the numbers rise from Monday (40) to Friday (90), so sales get larger as the week goes on. Friday is the busiest day, and Monday is the quietest. The pattern here is that sales increase across the week. From this, the shopkeeper can decide to order more drinks for Friday and can predict that weekends will probably be busy too. That is how statistics helps with a real decision.

1. The table below shows the favourite subject of 40 students in a school. | Subject | Number of students | |---------|--------------------| | Mathematics | 15 | | English | 10 | | Science | 10 | | History | 5 | | Total | 40 | (a) Which subject is most popular? (b) How many students chose either English or Science? (c) Why is it helpful to put this data in a table rather than leaving it as a long list of names?

answer: (a) Mathematics (15 students). (b) 10 + 10 = 20 students. (c) A table organises the data neatly so it is easy to read and compare at a glance — we can immediately see which subject is most popular without scanning a long list.

2. A farmer counts the number of eggs collected on five farms each morning and tabulates the results. | Farm | Eggs collected | |------|----------------| | A | 120 | | B | 95 | | C | 140 | | D | 110 | | E | 135 | (a) Which farm collected the fewest eggs? (b) What is the total number of eggs collected on all five farms? (c) If the farmer wants to know which farm is producing best, what does this table tell her? Use your own sentence to interpret it.

answer: (a) Farm B (95 eggs). (b) 120 + 95 + 140 + 110 + 135 = 600 eggs. (c) Farm C collected the most eggs (140) and Farm B the fewest (95), so Farm C is producing best and Farm B may need attention.

3. Write down two purposes of statistics, each explained in one sentence, and give a real-life example of how that purpose is used.

answer: Any two of the purposes with sensible examples, for example: To organise data so it is easy to read (a school lists its students' ages in a table); to spot patterns (a shop sees its sales rise each Friday); to make decisions and predictions (a farmer decides what to plant based on rainfall data); to make comparisons (comparing attendance in two terms).

Reading the Story in a Data Table

EXPLANATION

Once numbers are collected and written neatly into a table, the table can answer questions for you. Looking at a table to get information is called interpreting the data. When you interpret a table, you do two things: you read what is recorded, and you analyse it — that is, you compare the numbers, spot the biggest and smallest, and work out what the numbers are telling you.

A table is arranged in rows (going across) and columns (going down). Each row and each column is labelled so you know what the numbers mean. To find a value, trace along the row and down the column until they meet. To interpret, ask questions like:

Which category has the highest (or lowest) value?
How many more does one category have than another?
What is the total of all the categories?
What does the pattern of numbers suggest?

Here is an example. The table below shows how 40 students in a class travel to school.

<table>
<tr><th>Way of travelling</th><th>Number of students</th></tr>
<tr><td>Walking</td><td>18</td></tr>
<tr><td>Bus</td><td>12</td></tr>
<tr><td>Bicycle</td><td>7</td></tr>
<tr><td>Car</td><td>3</td></tr>
</table>

To interpret this table:

The most common way of travelling is walking, because 18 is the largest number.
The least common way is car, because 3 is the smallest number.
More students walk than go by bus: 18 − 12 = 6 more students.
The numbers add up: 18 + 12 + 7 + 3 = 40, which matches the 40 students mentioned.
The table tells us that walking is far more common than any motorised travel — this suggests many students live close to the school.

Remember: the numbers in a table are not just a list. They carry meaning, and your job is to read that meaning clearly.

WORKED_EXAMPLE

Worked example

The table below shows the number of books borrowed from a school library on five days of one week.

<table>
<tr><th>Day</th><th>Monday</th><th>Tuesday</th><th>Wednesday</th><th>Thursday</th><th>Friday</th></tr>
<tr><td>Books borrowed</td><td>25</td><td>30</td><td>28</td><td>35</td><td>42</td></tr>
</table>

a) Which day had the most books borrowed?
Look down the row for the largest number. The largest is 42, and that is under Friday. So Friday had the most books borrowed.

b) How many more books were borrowed on Friday than on Monday?
Friday had 42 and Monday had 25. Subtract: 42 − 25 = 17. So 17 more books were borrowed on Friday.

c) How many books were borrowed in the whole week?
Add all five numbers: 25 + 30 + 28 + 35 + 42 = 160. So 160 books were borrowed altogether that week.

d) What trend does the table show?
The numbers climb steadily from Monday (25) to Friday (42). This suggests that more students borrow books as the week goes on — perhaps to prepare for the weekend or an upcoming test.

1. A shop recorded the number of packets of biscuits it sold on each day of last week. <table> <tr><th>Day</th><th>Mon</th><th>Tue</th><th>Wed</th><th>Thu</th><th>Fri</th><th>Sat</th></tr> <tr><td>Packets sold</td><td>12</td><td>10</td><td>14</td><td>11</td><td>16</td><td>22</td></tr> </table> a) Which day sold the fewest packets? b) How many more packets were sold on Saturday than on Tuesday? c) How many packets were sold altogether in the week? d) Which two days were the shop's slowest, and what do the numbers suggest about them?

answer: a) Tuesday (10 packets). b) Saturday sold 22 and Tuesday sold 10, so 22 − 10 = 12 more packets. c) 12 + 10 + 14 + 11 + 16 + 22 = 85 packets. d) The two slowest days are Tuesday (10) and Monday (12). The numbers suggest the shop is quietest at the start of the week.

2. A class of 24 students was asked which of four fruits they like best. The results are in the table. <table> <tr><th>Fruit</th><th>Orange</th><th>Mango</th><th>Apple</th><th>Banana</th></tr> <tr><td>Number of students</td><td>9</td><td>8</td><td>4</td><td>3</td></tr> </table> a) Which fruit is liked by the most students? b) How many more students chose orange than banana? c) Add up the four numbers. Should the total match the number of students in the class? Explain. d) What does the table tell you about how popular mango is compared with the other fruits?

answer: a) Orange (9 students). b) 9 − 3 = 6 more students chose orange than banana. c) 9 + 8 + 4 + 3 = 24. Yes, the total should match the 24 students in the class, because every student chose one fruit. d) Mango is the second most popular fruit, chosen by 8 students — close to orange (9) and far more than apple (4) or banana (3).

C1 `The objective plus the lesson plan's design: position, method labels, materials, entry behaviour and assessment.`

Input — what the model was sent

Contract `ComposeWriteObjectiveSection`, declaring `artefact_spec, lesson_context, unit_context`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Need for statistic

`subtopic`

Need for statistics: purposes, and the collection and tabulation of data

`term`

Third Term

`week`

9

`lesson_objectives`

• explain the purpose of statistic
• analyse and interpret statistical data presented in tables

`entry_behaviour`

• Students are familiar with counting and recording information.

`instructional_materials`

• classroom register, bottle tops, pencils or counters.

Objective 1 — explain the purpose of statistic

`objective_text`

explain the purpose of statistic

`current_step`

1

`instructional_events`

• present_content

`assessment_points`

• (a) Define statistic

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, student_tasks, activities, source_evidence, source_figures, objective_learner_state`.

Objective 2 — analyse and interpret statistical data presented in tables

`objective_text`

analyse and interpret statistical data presented in tables

`current_step`

2

`instructional_events`

• provide_feedback

`assessment_points`

• (b) State three methods of collecting data.
• (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

Withheld from this payload: `step_title, step_content, instructional_type, estimated_minutes, student_tasks, activities, source_evidence, source_figures, objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

The Purpose of Statistics

EXPLANATION

Every day you gather information without really thinking about it. How many of your classmates were in school this morning? How many bottle tops did you and your friends collect last week? How many pencils are left in your pencil case? Whenever you count or record such facts, you are doing the first step of statistics.

Statistics is a method of collecting and organising information. Its purpose is to turn a jumble of facts and numbers into something you can understand and use. If you count the pupils in your class and simply say "the numbers were 30, 28, 32, 29," the information is there — but it is not yet organised. Statistics helps you arrange that information neatly so that you can see what is happening, compare things, and answer questions like "Which day had the most pupils present?" or "How many bottle tops do we have altogether?"

So statistics does two main jobs. First, it collects information by counting or recording facts. Second, it organises the information — often by putting it into a table — so that the facts become clear and easy to read. That is why statistics matters: it turns raw facts into useful knowledge.

WORKED_EXAMPLE

Let us see the two jobs of statistics in action.

Suppose a class keeps a register of who is present each day. At the end of the week the raw facts are:
Monday 30 pupils, Tuesday 28 pupils, Wednesday 32 pupils, Thursday 29 pupils, Friday 31 pupils.

These facts were collected by counting. Now statistics organises them by putting them into a table, like this:

Day   Number of pupils present
Monday   30
Tuesday   28
Wednesday   32
Thursday   29
Friday   31

Once the facts are organised, you can read them easily. Which day had the most pupils? Wednesday, with 32. The purpose of statistics is exactly this: collecting information and organising it into a table so that the facts become clear and useful.

1. Define statistics in your own words.

answer: Statistics is a method of collecting and organising information.

2. Your class counts the bottle tops you and your friends collected over four days. The raw facts are: Monday 15, Tuesday 12, Wednesday 18, Thursday 14. (a) Which of the two jobs of statistics was done by the counting? (b) Organise the facts into a table with the days in one column and the numbers in the other. (c) Using your table, which day had the most bottle tops?

answer: (a) Collecting the information. (b) Day | Bottle tops Monday | 15 Tuesday | 12 Wednesday | 18 Thursday | 14 (c) Wednesday, with 18.

Reading and Building a Frequency Table

EXPLANATION

When you collect information, you end up with a long list of numbers or words. A table helps you make sense of that list. One kind of table is a frequency table: it counts how many times each item appears, and it lets you see the pattern at a glance.

Here is the data everyone gave when asked how many siblings they have:

3, 2, 4, 3, 1, 2, 3, 4, 2, 3, 1, 3, 2, 4, 3, 1, 2, 3

That list is hard to read. To tidy it up, use tally marks. Go through the list one item at a time. For each "1" you meet, put one short line in the row for 1. When you reach the fifth mark in a row, cross it through so you can count in groups of five. Tallying the data above gives:

1: |||
2: |||||
3: |||| ||
4: |||

Count each group of marks. The count is the frequency — how many times that value appears. Now set it all out in a frequency table:

<table>
<tr><th>Number of siblings</th><th>Tally</th><th>Frequency</th></tr>
<tr><td>1</td><td>|||</td><td>3</td></tr>
<tr><td>2</td><td>|||||</td><td>5</td></tr>
<tr><td>3</td><td>|||| ||</td><td>7</td></tr>
<tr><td>4</td><td>|||</td><td>3</td></tr>
<tr><td>Total</td><td></td><td>18</td></tr>
</table>

The frequency column now shows the whole picture: the most common number of siblings is 3, because 7 students chose it, and 1 and 4 each appear 3 times. Add the frequencies together to check your table — they must equal the number of items you started with (here 3 + 5 + 7 + 3 = 18).

1. The ages (in years) of the students in a club are: 11, 12, 11, 13, 12, 11, 12, 13, 11, 12, 13, 12, 11, 12. (a) Tally the data and set it out in a frequency table. (b) How many students are in the club? (c) Which age appears most often?

answer: (a) Frequency table:
11: 5
12: 6
13: 3
Total: 14
(b) 14 students
(c) Age 12 appears most often.

C2 `C1 plus the lesson note as source evidence.`

Input — what the model was sent

Contract `ComposeWriteGroundedSection`, declaring `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Need for statistic

`subtopic`

Need for statistics: purposes, and the collection and tabulation of data

`term`

Third Term

`week`

9

`lesson_objectives`

• explain the purpose of statistic
• analyse and interpret statistical data presented in tables

`entry_behaviour`

• Students are familiar with counting and recording information.

`instructional_materials`

• classroom register, bottle tops, pencils or counters.

Objective 1 — explain the purpose of statistic

objective_text

explain the purpose of statistic

current_step

1

instructional_events

• present_content

assessment_points

• (a) Define statistic

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "statistics is the science of collecting, organizing, presenting, analyzing and
interpreting\ndata.\n\n#### Meaning of Data\n\nData are facts or information collected for a
particular purpose.\n\n#### Importance of statistics\n\nIt helps us to\n\n- (a) Collect
Information\n- (b) organize Information\n- (c) Compare Information\n- (d) Make decisions\n- (e)
Present Information Clearly."
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 2 — analyse and interpret statistical data presented in tables

`objective_text`

analyse and interpret statistical data presented in tables

`current_step`

2

`instructional_events`

• provide_feedback

`assessment_points`

• (b) State three methods of collecting data.
• (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

`lesson_content`


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Collection of Data\n\nData can be collected through\n\n1. observation\n2.
counting\n3. Interview\n4. Questionnaire.\n\n#### Tally Marks\n\nTally Marks are used to record
data as they are collected.\n\n**Example:** The fig below show the same data collected in two
different ways by two student\n\n(a) bus, car, car, car, lorry, bicycle, bicycle, car, car,
lorry, bicycle, car.\n\nThe above data can be represented in Tally\n\n| Vehicles | Tally | Total
|\n|---|---|---|\n| car | \\|\\|\\|\\|\\|/ \\| | 6 |\n| bus | \\| | 1 |\n| lorry | \\|\\| | 2 |\n|
bicycle | \\|\\|\\| | 3 |\n| Motor Bike | | |\n\n#### Frequency Table\n\nA frequency table is a
table showing how many times each item occurs in a set of data.\n\n**Examples:**\n\nFavourite
colours of student:\n\n| colour | frequency |\n|---|---|\n| Red | 18 |\n| Blue | 20 |\n| Green |
14 |\n| Yellow | 16 |\n| White | 15 |\n| Pink | 30 |"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

What Statistics Is and Why We Need It

EXPLANATION

Every day you come across facts and numbers: how many classmates were present, how many goals your team scored, how much money you spent on snacks. Facts or information like these, collected for a particular purpose, are called data.

When you have a pile of data, you cannot make much sense of it just by staring at it. Statistics is the science of collecting, organising, presenting, analysing and interpreting data. It is the set of steps that turns a jumble of facts into something you can understand and use.

Why bother with all this? Statistics helps us to:

Collect information — gather the facts we need in the first place.
Organise information — arrange data so it is easy to read, for example by putting it in a table.
Compare information — see how one set of facts stacks up against another.
Make decisions — use what the data tells us to choose what to do next.
Present information clearly — show the facts in a way that others can follow easily.

So statistics is not just about numbers — it is about giving those numbers a purpose. From a simple count of who is present in class to the results of a whole country's census, statistics is what lets us turn raw data into useful knowledge.

WORKED_EXAMPLE

Worked example. Suppose your class wants to know how many boys and how many girls are present today. One person calls each name while another ticks a column marked "Boys" or "Girls" on the register.

At the end you have a set of facts — the ticks on the register. That is your data: information collected for the purpose of finding the class attendance.

Now you organise that data. You count the ticks in each column and write the two totals side by side in a small table:

<table>
<tr><th>Boys</th><th>Girls</th></tr>
<tr><td>12</td><td>14</td></tr>
</table>

With the data organised, you can compare the two numbers — you can see there are more girls present than boys. The totals can then be used to make a decision, such as whether there are enough desks for everyone present. Finally, you present the result clearly by showing the table to the class.

Every one of those steps — collecting, organising, comparing, deciding and presenting — is statistics at work on real data.

1. In your own words, what is statistics?
answer: Statistics is the science of collecting, organising, presenting, analysing and interpreting data.

2. What is meant by the word "data"?
answer: Data are facts or information collected for a particular purpose.

3. Give three ways that statistics helps us, and briefly explain each in your own words.
answer: Any three of: collect information (gather the facts we need), organise information (arrange data so it is easy to read), compare information (see how one set of facts stacks up against another), make decisions (use the data to choose what to do), or present information clearly (show the facts so others can follow easily).

Collecting, Tallying and Tabulating Data

EXPLANATION

Before you can make sense of information, you first have to gather it. Information that is collected is called **data**, and there are several ways to collect it:

1. **Observation** — you watch something happen and note what you see.
2. **Counting** — you count how many of each thing there is.
3. **Interview** — you ask people questions and listen to their answers.
4. **Questionnaire** — you give people a set of written questions to answer.

Once you have collected data, you need a neat way to record it as you go. **Tally marks** do exactly that. To make a tally, you draw one small stroke (|) for each item you count, and every fifth item you draw a stroke across the four you have already drawn, making a group of five: ||||/

Let's see this with some vehicles. Suppose these were collected by counting the vehicles passing by:

bus, car, car, car, lorry, bicycle, bicycle, car, car, lorry, bicycle, car

To record them with tally marks, you go through the list one at a time and draw a stroke for each. The cars come up six times, so they are shown as ||||/ plus one more stroke (||||/ |). The bus comes up once, shown as |. The lorry comes up twice, shown as ||. The bicycle comes up three times, shown as |||. The total column shows the count for each vehicle.

EXPLANATION

When your tally marks are complete, you can tidy them into a **frequency table**. A frequency table is a table showing how many times each item occurs in a set of data. The number of times an item occurs is called its **frequency**.

Here is the vehicles data set out in a frequency table:

Vehicles	Tally	Total
car	||||/ |	6
bus	|	1
lorry	||	2
bicycle	|||	3

Each total comes straight from counting the tally marks in that row, and the totals should add up to the whole set of data. Here, 6 + 1 + 2 + 3 = 12, which is exactly how many vehicles were in the list.

A frequency table can also just show each item with its frequency, without the tally column. Here is one showing the favourite colours of some students:

colour	frequency
Red	18
Blue	20
Green	14
Yellow	16
White	15
Pink	30

Reading a frequency table means looking up an item and reading across to find how many times it occurred. You can also add the frequencies to find the total number of items in the data, or compare frequencies to see which item occurred most or least often.

WORKED_EXAMPLE

Worked example — building a frequency table with tally marks

Here is a set of numbers: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2

> Follow each step to turn this data into a frequency table.
>
> **Step 1: Collect the data.** The data is already listed above — 13 numbers in all.
>
> **Step 2: Record with tally marks.** Go through the list one number at a time and draw a stroke for each. The number 1 appears three times, so it gets |||. The number 2 appears three times, so it gets |||. The number 3 appears four times, so it gets ||||/. The number 4 appears three times, so it gets |||.
>
> **Step 3: Set it out in a frequency table.**
>
> | Number | Tally | Frequency |
> |---|---|---|
> | 1 | \|\|\| | 3 |
> | 2 | \|\|\| | 3 |
> | 3 | \|\|\|/ | 4 |
> | 4 | \|\|\| | 3 |
>
> **Step 4: Check your work.** Add the frequencies: 3 + 3 + 4 + 3 = 13, which matches the 13 numbers in the data. The frequency table is correct.

1. Use tally marks to record these vehicles, then draw up a frequency table showing each vehicle's frequency. car, bus, car, car, lorry, bicycle, bicycle, car, car, lorry, bicycle, car How many of each vehicle are there, and how many vehicles were collected in total?

answer: car: 6, bus: 1, lorry: 2, bicycle: 3. Total = 12 vehicles.

2. Here is a frequency table of the favourite colours of a class of students: | colour | frequency | |---|---| | Red | 18 | | Blue | 20 | | Green | 14 | | Yellow | 16 | | White | 15 | | Pink | 30 | (a) Which colour was chosen by the most students, and how many chose it? (b) Which colour was chosen by the fewest students, and how many chose it? (c) How many students took part altogether?

answer: The most popular colour is Pink, with a frequency of 30. The least popular colour is Green, with a frequency of 14. Altogether, 18 + 20 + 14 + 16 + 15 + 30 = 113 students took part.

C3 `C2 plus a validation and repair pass over the draft.`

Input — what the model was sent

Contract `ComposeWriteGroundedSection`, declaring `artefact_spec, lesson_context, unit_context, lesson_content`.

Lesson context — shared by every section

`subject`

Mathematics

`grade_level`

JSS1

`topic`

Need for statistic

`subtopic`

Need for statistics: purposes, and the collection and tabulation of data

`term`

Third Term

`week`

9

`lesson_objectives`

• explain the purpose of statistic
• analyse and interpret statistical data presented in tables

`entry_behaviour`

• Students are familiar with counting and recording information.

`instructional_materials`

• classroom register, bottle tops, pencils or counters.

Objective 1 — explain the purpose of statistic

objective_text

explain the purpose of statistic

current_step

1

instructional_events

• present_content

assessment_points

• (a) Define statistic

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "statistics is the science of collecting, organizing, presenting, analyzing and
interpreting\ndata.\n\n#### Meaning of Data\n\nData are facts or information collected for a
particular purpose.\n\n#### Importance of statistics\n\nIt helps us to\n\n- (a) Collect
Information\n- (b) organize Information\n- (c) Compare Information\n- (d) Make decisions\n- (e)
Present Information Clearly."
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Objective 2 — analyse and interpret statistical data presented in tables

objective_text

analyse and interpret statistical data presented in tables

current_step

2

instructional_events

• provide_feedback

assessment_points

• (b) State three methods of collecting data.
• (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

lesson_content


```
[
 {
  "source_ref": "the lesson note",
  "excerpt": "#### Collection of Data\n\nData can be collected through\n\n1. observation\n2.
counting\n3. Interview\n4. Questionnaire.\n\n#### Tally Marks\n\nTally Marks are used to record
data as they are collected.\n\n**Example:** The fig below show the same data collected in two
different ways by two student\n\n(a) bus, car, car, car, lorry, bicycle, bicycle, car, car,
lorry, bicycle, car.\n\nThe above data can be represented in Tally\n\n| Vehicles | Tally | Total
|\n|---|---|---|\n| car | \\|\\|\\|\\|/ \\| | 6 |\n| bus | \\| | 1 |\n| lorry | \\|\\| | 2 |\n|
bicycle | \\|\\|\\| | 3 |\n| Motor Bike | | |\n\n#### Frequency Table\n\nA frequency table is a
table showing how many times each item occurs in a set of data.\n\n**Examples:**\n\nFavourite
colours of student:\n\n| colour | frequency |\n|---|---|\n| Red | 18 |\n| Blue | 20 |\n| Green |
14 |\n| Yellow | 16 |\n| White | 15 |\n| Pink | 30 |"
 }
]
```


Withheld from this payload: `step_title`, `step_content`, `instructional_type`, `estimated_minutes`, `student_tasks`, `activities`, `source_evidence`, `source_figures`, `objective_learner_state`.

Output — the pack it produced (draw 3)

Objective 1

What Statistics Is For

EXPLANATION

Before you study anything in statistics, it helps to know what the word actually means. Statistics is the science of collecting, organising, presenting, analysing and interpreting data. In other words, statistics is the method you use to take raw information and turn it into something you can understand and use.

So what is data? Data are facts or information collected for a particular purpose. The word "data" is just another name for the pieces of information you gather. When you count the number of bottles on a shelf, record the ages of your classmates, or write down how many pencils each pupil has, you are collecting data.

Because you are already used to counting and recording information, you already have a start on what statistics asks of you. Statistics just takes that counting and recording and gives it a clear purpose.

EXPLANATION

Statistics matters because it helps you do five important things with information:

- Collect information — gather the facts you need.
- Organise information — sort and arrange those facts so they are easy to work with.
- Compare information — look at one set of facts against another.
- Make decisions — use the facts to choose sensibly.
- Present information clearly — show the facts so others can understand them too.

Each of these is a purpose of statistics. Whether you are counting bottle tops in a store, recording the number of pupils present in each class, or comparing scores in a test, you are using statistics to collect, organise, compare, decide and present.

WORKED_EXAMPLE

Worked example

Imagine your class wants to know how many pencils each pupil has. Let's see how the purposes of statistics apply to this one small task.

First, you collect the information — you ask each pupil and write down the number of pencils they have. That is collecting information.

Next, you organise the information — you arrange the numbers, perhaps from smallest to largest, so you can see them clearly.

Then you can compare the information — you can say one pupil has more pencils than another, or that most pupils have about the same number.

Finally you present the information clearly — you write it in a table or list so anyone can read it — and use it to make a decision, such as which pupil might need more pencils.

So collecting, organising, comparing, deciding and presenting are not five separate ideas floating on their own. They are the steps you move through whenever you work with data.

1. State what the word "data" means in statistics.
answer: Data are facts or information collected for a particular purpose.

2. List the five things statistics helps us to do with information.
answer: Collect information, organise information, compare information, make decisions, and present information clearly.

3. Define statistics.
answer: Statistics is the science of collecting, organising, presenting, analysing and interpreting data.

Collecting, Tallying and Tabulating Data

EXPLANATION

Statistics is all about making sense of numbers, and before you can analyse or interpret anything you first have to gather the information. The facts and figures you collect are called data. Data can be collected in four main ways:

Observation — you watch and record what you see happening. For example, sitting by the roadside and noting the types of vehicles that pass.
Counting — you count how many of something there are. For example, counting the number of pupils in your class.
Interview — you ask people questions and record their answers face to face.
Questionnaire — you give people a set of written questions to answer.

Once you have collected your data, you need a tidy way to record it as it comes in. That is where tally marks come in.

EXPLANATION

A tally mark is a single stroke made for each item you see or count. Every time an item appears, you add one stroke. To make the strokes easy to count in fives, you draw the fifth stroke across the previous four, like this: |||| — five strokes in one group.

Suppose you recorded the vehicles passing your house as:
bus, car, car, car, lorry, bicycle, bicycle, car, car, lorry, bicycle, car.

You can count the cars as you go: car, car, car, car, car, car — that is six cars. Buses appear once, lorries twice, and bicycles three times. The same data can be set out in a tally table like this:

<table>
<tr><th>Vehicles</th><th>Tally</th><th>Total</th></tr>
<tr><td>car</td><td>||||/ |</td><td>6</td></tr>
<tr><td>bus</td><td>|</td><td>1</td></tr>
<tr><td>lorry</td><td>||</td><td>2</td></tr>
<tr><td>bicycle</td><td>|||</td><td>3</td></tr>
</table>

The tally column shows how the counting was done, and the total column shows how many of each item there are.

EXPLANATION

A tally table that shows how many times each item occurs is called a frequency table. A frequency table is simply a table showing how many times each item occurs in a set of data. The number of times an item appears is its frequency.

Here is a frequency table for the favourite colours of a group of students:

<table>
<tr><th>Colour</th><th>Frequency</th></tr>
<tr><td>Red</td><td>18</td></tr>
<tr><td>Blue</td><td>20</td></tr>
<tr><td>Green</td><td>14</td></tr>
<tr><td>Yellow</td><td>16</td></tr>
<tr><td>White</td><td>15</td></tr>
<tr><td>Pink</td><td>30</td></tr>
</table>

Because the data is now arranged in a table, you can read it at a glance — for example, you can see at once that Pink was the most popular colour (30 students) and Green was the least popular (14 students). That is what it means to analyse and interpret data that is set out in a table.

WORKED_EXAMPLE
Worked example — building a frequency table

Here is a set of data collected by counting the scores pupils got when asked to name a shape:
1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2

Step 1: Look at each number in turn and make a tally mark for it. Start with the first number, 1: add one stroke. Then the second number, 1: add another stroke. Then 3, then 4, and so on.

Step 2: When you have gone through every number, group the tally marks in fives. You should get:

1 appears three times: |||
2 appears three times: |||
3 appears four times: ||||
4 appears three times: |||

Step 3: Set the results out in a frequency table.

<table>
<tr><th>Score</th><th>Tally</th><th>Frequency</th></tr>
<tr><td>1</td><td>|||</td><td>3</td></tr>
<tr><td>2</td><td>|||</td><td>3</td></tr>
<tr><td>3</td><td>||||</td><td>4</td></tr>
<tr><td>4</td><td>|||</td><td>3</td></tr>
</table>

To check your work, add up all the frequencies: 3 + 3 + 4 + 3 = 13, which matches the 13 numbers in the data. If the totals match, your table is correct.

1. List the four ways in which data can be collected.
answer: Observation, counting, interview and questionnaire.

2. A pupil recorded the fruit sold at a stall for a morning: mango, mango, apple, orange, mango, orange, apple, mango, orange, mango, apple, orange, mango, apple, mango. Prepare a frequency table for this data.
answer: Mango: 7, Apple: 4, Orange: 4.

3. The frequency table below shows the favourite sports of a group of students. <table><tr><th>Sport</th><th>Frequency</th></tr><tr><td>Football</td><td>22</td></tr><tr><td>Swimming</td><td>12</td></tr><tr><td>Basketball</td><td>15</td></tr><tr><td>Running</td><td>11</td></tr></table> (a) How many students chose basketball? (b) Which sport was the most popular? (c) How many students were asked altogether?
answer: (a) 15. (b) Football (22). (c) 22 + 12 + 15 + 11 = 60 students.

4. Here is a set of data collected by counting: 2, 5, 2, 3, 5, 2, 3, 5, 2, 3. Prepare a frequency table for this data.
answer: 2: 4, 3: 3, 5: 3.

Result — per rubric dimension

Scores are 1–5. Every figure is the mean of 3 draws rated by 2 judge pass(es) — 6 judgments per cell.

Rubric	C0	C1	C2	C3	C1 – C0	C2 – C1	C3 – C2
Objective and curriculum alignment	4.00	5.00	5.00	5.00	**+1.00**	+0.00	+0.00
Taught-lesson alignment	3.00	4.83	5.00	5.00	**+1.83**	**+0.17**	+0.00
Scope and conceptual correctness	5.00	5.00	5.00	5.00	+0.00	+0.00	+0.00
Student-facing clarity and usefulness	5.00	5.00	5.00	5.00	+0.00	+0.00	+0.00

Why the judge scored it that way

Every rationale the judge wrote for this lesson, deduplicated across draws and passes. These are its words, not a summary.

Objective and curriculum alignment — C0 4.00, C1 5.00, C2 5.00, C3 5.00

- `C0` Both lesson objectives are substantially covered, and the curriculum target for JSS1 statistics is well represented. However, the pack never asks students to construct a frequency table from raw data (tally marks to frequency table), which is part of the stated objectives and the teacher's own evaluation, so full marks are not warranted.
- `C0` Both lesson objectives are addressed and the curriculum target for JSS1 statistics is well served, but the specific skill of constructing a frequency table using tally marks (the method the teacher demonstrated and set in the evaluation) is absent, leaving a gap in full objective coverage.
- `C0` Both lesson objectives are clearly addressed with appropriate depth for JSS1. The only gap is that tally marks and frequency table construction — which appear in the teacher's evaluation question 3(c) — are not practised in the pack; the pack focuses on reading ready-made tables rather than building them.
- `C0` Both lesson objectives are addressed meaningfully, but the pack stops short of the full curriculum target by omitting the construction of a frequency table (including tally marks) from raw data, which was explicitly part of the lesson and the evaluation.
- `C0` The pack covers both stated objectives reasonably well, with solid treatment of purposes and table interpretation. However, the key skill of constructing a frequency table with tally marks (directly tested in the teacher's evaluation) is absent, which limits full alignment.
- `C0` Both lesson objectives are substantively addressed, and the curriculum topic of data collection, tabulation, and interpretation is well represented. The one gap is the absence of any activity requiring students to construct a frequency table from raw data, which was explicitly part of the teacher's evaluation.
- `C1` Both lesson objectives are explicitly and fully covered: the purpose of statistics and the analysis/interpretation of data in tables. The pack maps precisely onto the stated curriculum targets for JSS1 Term 3, Week 9.
- `C1` Both lesson objectives are addressed directly and completely. The pack maps onto the stated curriculum target (purposes of statistics, data collection and tabulation) without any gaps.
- `C1` Both lesson objectives are fully covered in the revision pack — purpose of statistics and construction/interpretation of frequency tables — with practice questions that closely match the teacher's own evaluation questions. Alignment is thorough.
- `C1` Both lesson objectives are fully and explicitly covered. The pack maps directly onto the stated curriculum targets for JSS1 Term 3 Week 9, including the specific data set used in the teacher's evaluation.
- `C1` Both lesson objectives are covered fully and explicitly. The pack maps one-to-one onto the stated objectives and the curriculum content for JSS1 statistics, with no gaps.
- `C1` Both lesson objectives are fully covered in the revision pack, and the practice questions align closely with the teacher's own evaluation questions, confirming strong curriculum alignment.
- `C2` Both stated lesson objectives are fully and explicitly covered. The pack addresses the definition, purposes, data collection methods, tally marks, and frequency tables — everything the curriculum target requires for this topic at JSS1.
- `C2` Both lesson objectives are fully covered, and the pack maps precisely onto the curriculum target for JSS1 Term 3 Week 9. Nothing important is missing.

- `C2` Both lesson objectives are fully covered in the correct order. The pack addresses the definition and purpose of statistics, then moves to collecting, tallying, and tabulating data into frequency tables, matching the curriculum targets exactly.
- `C2` The pack covers both stated objectives completely and mirrors all three of the teacher's own evaluation questions, demonstrating tight alignment with the intended curriculum targets for this lesson.
- `C2` Both lesson objectives are fully and explicitly covered. The pack maps cleanly onto the curriculum target of data collection, tabulation, and interpretation at JSS1 level.
- `C2` Both lesson objectives are fully covered with appropriate depth, and the pack maps directly onto the teacher's stated evaluation questions, confirming strong curriculum alignment.
- `C3` Both lesson objectives are fully and explicitly covered. The pack maps onto the curriculum targets with no gaps, giving JSS1 students everything the lesson intended them to learn.
- `C3` Both lesson objectives are fully covered with appropriate depth, and the pack maps precisely onto the teacher's stated evaluation questions, leaving no objective unaddressed.
- `C3` Both stated lesson objectives are fully and directly addressed. The pack covers defining statistics, its purposes, methods of data collection, tally marks, and frequency tables — all aligned with the JSS1 Third Term statistics topic.
- `C3` Both stated objectives are fully addressed in the correct order, and the pack's content maps directly onto the JSS1 Third Term statistics curriculum, including data collection methods and frequency tables.
- `C3` Both lesson objectives are fully and explicitly covered, and the pack even reproduces the teacher's exact evaluation questions with complete worked solutions. Curriculum alignment is thorough.
- `C3` Both lesson objectives are explicitly and thoroughly covered, and the pack maps cleanly onto the curriculum target of understanding the need for statistics and working with frequency tables. Nothing important from the stated objectives is missing.

Taught-lesson alignment — C0 3.00, C1 4.83, C2 5.00, C3 5.00

- `C0` The pack aligns with the conceptual introduction part of the lesson but omits the core pedagogical activity (tally marks and constructing a frequency table from raw data) that the teacher planned and assessed. It covers table-reading more deeply than the lesson seems to have required, while missing the construction side.
- `C0` The pack follows the lesson's broad sequence and covers the conceptual introduction well, but it omits the tally-mark recording step and the construction of frequency tables from raw data, which were the practical heart of the lesson and the method used in evaluation.
- `C0` The pack aligns well with the conceptual sequence and the reading-tables strand, but it omits the hands-on data-collection and tally/frequency-table-building activity that formed the second half of the taught lesson, leaving a meaningful gap in taught-lesson coverage.
- `C0` The pack captures the conceptual framing of the lesson but misses the central practical activity — building a frequency table using tally marks from a raw data list — which was the main student task and the teacher's own evaluation question; alignment is partial rather than full.
- `C0` While the pack follows the broad topic area, it omits the lesson's central taught activity — recording with tally marks and producing a frequency table — and introduces emphasis on purposes that the lesson treated briefly, making the alignment only partial.
- `C0` The pack follows the broad lesson sequence and covers interpretation of tables as the lesson intended, but it omits the tally-mark recording method that was a specific instructional focus, and does not replicate the participatory data-collection activity that characterised the lesson.
- `C1` The pack faithfully mirrors the lesson's sequence, emphasis, and activities, including the specific examples and materials the teacher used, and reproduces the teacher's own assessment task at the appropriate depth.
- `C1` The pack mirrors the lesson's sequence, method (tally → frequency table) and expected depth very closely. The only gap is the absence of any reference to the hands-on, object-counting activity that anchored the lesson's introduction, which slightly reduces alignment to what was actually taught.
- `C1` The pack mirrors the lesson's order, emphasis, and chosen activities (register, bottle tops, tally marks, frequency table) almost exactly. It reflects the taught lesson faithfully and at the expected depth for JSS1 Third Term week 9.
- `C1` The revision pack mirrors the lesson's order, emphasis, and activities closely. The depth matches what a JSS1 lesson introduction to statistics would require, and the teacher's own examples and materials are woven throughout.
- `C1` The revision pack faithfully reflects the lesson's sequence, method, and expected depth. The use of tally marks, the same data set, and a matching depth of interpretation questions show strong alignment with what was taught.

- `C1` The revision pack mirrors the lesson's sequence, emphasis, and activities very closely, including the use of the same concrete materials referenced by the teacher, reflecting the taught lesson faithfully.
- `C2` The pack mirrors the lesson's two-part structure and depth faithfully, moving from concept introduction to hands-on tabulation in the same order the teacher planned. The depth — defining terms, listing purposes, building tally/frequency tables — matches what a JSS1 lesson of this type would cover.
- `C2` The revision pack mirrors the lesson's sequence, emphasis, concrete materials, and expected depth throughout, making it a faithful reflection of what was taught.
- `C2` The revision pack mirrors the lesson's order, emphasis, and activities closely, including the register-based introduction and the identical data set from the teacher's evaluation. The depth expected at JSS1 is well matched.
- `C2` The pack reproduces the lesson's sequence, method (tally then frequency table), and expected depth with high fidelity, and even uses the teacher's own data set as the central worked example, showing strong alignment to how the lesson was actually taught.
- `C2` The revision pack mirrors the lesson's order of presentation, its examples, and its expected depth faithfully. A student who attended the lesson would find this a natural continuation of what was taught.
- `C2` The pack mirrors the teacher's instructional sequence, chosen methods (tally marks, frequency tables), and the specific data used in evaluation, showing close alignment with what was taught and how it was taught.
- `C3` The pack preserves the lesson's sequence, emphasis, and method. A student revising at home encounters the same conceptual journey and level of detail as the classroom lesson, making it an accurate and faithful reflection of what was taught.
- `C3` The pack faithfully reproduces the lesson's sequence, key activities (tallying and tabulating), and the specific data set used by the teacher, reflecting very close alignment with how the lesson was taught.
- `C3` The pack closely mirrors the lesson's sequence, depth, and activities. It does not skip ahead or introduce topics beyond what the teacher planned, and the frequency-table building exercise replicates the teacher's own evaluation task.
- `C3` The pack faithfully reflects the taught lesson's sequence, emphasis, and activities, from introducing statistics as organised information through to building a frequency table from raw data, at exactly the depth a JSS1 lesson would reach.
- `C3` The pack mirrors the lesson's order, method, and depth faithfully — from introducing statistics as a concept through to building frequency tables — with no notable gaps or mismatches.
- `C3` The revision pack closely reflects the lesson's sequence, emphasis, and instructional approach, including the same worked dataset the teacher planned to use in evaluation, making it a faithful take-home mirror of what was taught.

Scope and conceptual correctness — C0 5.00, C1 5.00, C2 5.00, C3 5.00

- `C0` Every numerical calculation and conceptual statement checked is correct, and the content remains firmly within the JSS1 statistics boundary. No errors or out-of-scope content were found.
- `C0` There are no mathematical or conceptual errors anywhere in the pack, and the content is appropriately scoped for JSS1; nothing is introduced that is beyond the lesson boundary.
- `C0` There are no conceptual or computational errors, and the content stays firmly within the boundaries of the planned lesson without straying into topics not yet taught.
- `C0` There are no mathematical or conceptual errors anywhere in the pack, and it stays appropriately within the scope of an introductory JSS1 statistics lesson without over-reaching.
- `C0` No conceptual or computational errors were found anywhere in the pack; every definition, worked example, and answer key item is factually correct and appropriately scoped for JSS1.
- `C0` There are no mathematical or conceptual errors anywhere in the pack, and it remains well within the scope expected for a JSS1 introductory statistics lesson.
- `C1` The pack stays strictly within JSS1 scope — basic data collection, tally marks, and frequency tables — and contains no conceptual or computational errors anywhere in the content or the answer key.
- `C1` All numerical work is accurate and the check-sum verification technique is mathematically sound. The content stays strictly within the expected JSS1 introductory statistics boundary.
- `C1` All factual claims, definitions, tallies, frequencies, and totals in the pack are accurate. The content stays within the JSS1 scope — no advanced statistical concepts are introduced — and there are no mathematical errors.
- `C1` All tables, tallies, frequencies, and totals are arithmetically correct, and the pack stays well within the JSS1 scope — no concepts beyond what was taught are introduced.
- `C1` Every figure in the pack is mathematically correct, and the content remains firmly within the expected lesson boundary without introducing out-of-scope concepts.

- `C1` All mathematical content is accurate, the scope is appropriate for JSS1, and no errors or out-of-scope material were found.
- `C2` No conceptual or mathematical errors were found anywhere in the pack. The scope stays within what is expected for this JSS1 statistics introduction — no overreach into measures of central tendency or graphs, which would be beyond this lesson.
- `C2` The pack stays within the lesson boundary — no topics beyond what was taught are introduced — and every fact, definition, and calculation is conceptually and mathematically correct.
- `C2` There are no conceptual or mathematical errors anywhere in the pack. The scope stays squarely within JSS1 introductory statistics — no material is introduced that goes beyond what the lesson planned.
- `C2` Every calculation and definition in the pack is conceptually correct and appropriately scoped; the content stays strictly within what a JSS1 week-9 statistics introduction should cover.
- `C2` The pack stays squarely within JSS1 scope with no introduced errors, no over-extension into topics not yet taught, and all mathematical and conceptual content is correct.
- `C2` The pack stays within the expected JSS1 lesson boundary without introducing out-of-scope concepts, and all numerical work and definitions are conceptually and mathematically correct.
- `C3` No conceptual or mathematical errors are present, and the content stays firmly within the JSS1 scope. Nothing is introduced that goes beyond what the lesson plan requires.
- `C3` All definitions, methods, worked examples, and answers are conceptually and mathematically correct, and no content exceeds or falls short of JSS1 expectations for this topic.
- `C3` The pack stays firmly within JSS1 scope — no advanced statistics concepts are introduced. All definitions, tallies, frequency counts, and answer-key values are mathematically correct with no errors detected.
- `C3` Every concept and calculation in the pack is correct, the check-sums are all verified, and the content does not stray outside the intended lesson boundary.
- `C3` All definitions, tallying conventions and frequency counts are accurate, and the pack stays strictly within the boundaries of the taught lesson without introducing premature concepts.
- `C3` The pack stays firmly within JSS1 scope and every fact, count, and definition has been checked and is correct; there are no mathematical or conceptual errors anywhere in the material.

Student-facing clarity and usefulness — C0 5.00, C1 5.00, C2 5.00, C3 5.00

- `C0` The pack is exceptionally well structured for independent home use: clear explanations, model solutions with visible steps, varied practice questions, and a detailed answer key all combine to make it highly usable for a JSS1 learner without a teacher present.
- `C0` The pack is very well structured for independent home use: clear headings, relatable contexts (eggs, drinks, school attendance), worked examples before practice, and detailed answers mean a JSS1 student can work through it without a teacher present.
- `C0` The pack is exceptionally well-structured for independent home use: clear explanations, approachable worked examples, varied practice questions, and thorough answer guidance make it highly useful for a JSS1 student revising alone.
- `C0` The pack is exceptionally well-structured for independent home use: clear explanations, scaffolded examples, varied practice questions, and a detailed answer key all make it highly useful and accessible to a JSS1 student working alone.
- `C0` The pack is exceptionally well-structured for independent home use: language is simple and age-appropriate, examples are contextually familiar (drinks, eggs, school attendance), and the answer section gives enough working for genuine self-correction.
- `C0` The pack is very well structured for independent home use: explanations are clear, worked examples model exactly what students must do, and the answer key allows self-checking with understanding rather than just answer-matching.
- `C1` The pack is exceptionally well-suited as independent take-home material: the writing level, relatable examples, structured steps, and detailed answer key give a JSS1 learner everything needed to revise and self-assess without adult guidance.
- `C1` The pack is exceptionally well structured for unsupported home use: plain language, scaffolded steps, contextualised examples and fully explained answers give a JSS1 student everything needed to review and practise without a teacher present.
- `C1` The pack is written at an accessible level for JSS1 learners, with relatable contexts, a clear worked example, structured practice questions, and a self-marking answer key — making it genuinely useful as a standalone take-home revision resource.
- `C1` The pack is exceptionally well-structured for independent home use: clear explanations, worked examples, practice questions, and fully explained solutions give a JSS1 student everything needed to revise without teacher support.
- `C1` The pack is well-structured, uses accessible language, and provides everything a JSS1 student needs to revise independently — clear explanations, worked examples, practice questions, and model answers.

- **C1** The pack is written in accessible language with relatable examples and full worked solutions, making it highly suitable for independent home revision by a JSS1 learner.
- **C2** The pack is exceptionally well-suited as a take-home revision tool: it explains concepts clearly, models the key skill step-by-step, provides practice questions that build from simple recall to table construction, and supplies complete answers for self-marking.
- **C2** The pack is well-structured, uses accessible language appropriate for JSS1, and provides enough scaffolding and self-check support for a student to use it meaningfully without a teacher present.
- **C2** The pack is well-structured, uses plain English appropriate for JSS1 learners, and provides enough guidance through worked examples and a full answer key that a student can work through it independently and meaningfully at home.
- **C2** The pack is exceptionally well structured for independent home use: vocabulary is introduced gently, the worked example is thorough and clearly staged, and the answer section allows students to check and understand their work without a teacher present.
- **C2** The language is accessible for JSS1 learners, the structure is logical and self-contained, and the answer key with explanations makes it genuinely usable as independent take-home revision.
- **C2** The pack is written at an accessible level for JSS1 learners, uses familiar contexts, and provides enough scaffolding and answer checking that a student can work through it independently and verify their own understanding.
- **C3** The pack is well-structured, clearly written, and self-contained. A JSS1 student working alone at home can read the explanations, follow the worked examples, attempt the practice questions, and check their own answers without any additional support.
- **C3** The pack is exceptionally well-structured for independent home use: relatable examples, scaffolded worked examples, and a thorough answer key with explanations make it highly accessible and useful for a JSS1 learner.
- **C3** The pack is exceptionally well suited for independent home use: language is simple and encouraging, worked examples scaffold the tasks, practice questions graduate in difficulty, and the answer key is thorough enough for self-correction.
- **C3** The pack is exceptionally well-structured for independent home use: concepts are explained plainly, worked examples scaffold each skill, and full annotated answers allow a JSS1 student to check and understand their own work.
- **C3** The language is plain, the structure is logical, and the self-check answers are detailed enough for an independent learner at JSS1 level to use the pack meaningfully without a teacher present.
- **C3** The pack is exceptionally well-organised for independent home use: the explanations are accessible, the worked examples model exactly what students are asked to do, and the answer key enables effective self-assessment without a teacher present.